

THENMOZHI SEKAR

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About Me: I am an AI/ML Research Engineer with 4.5+ years of experience in machine learning, Generative AI, and scalable AI systems. Skilled in developing AI-powered applications, LLM solutions, PyTorch-based models, and production-grade ML pipelines, with published research in IEEE venues

EDUCATION

Quinnipiac University,
Master of Science in Computer Science, **CGPA – 3.8/4**
Madras University,
Bachelor of Computer Applications, **CGPA - 3.7/4**

Hamden, CT, USA
Aug 2024 – May 2026
Kanchipuram, TN, India
Aug 2018 – May 2021

EXPERIENCE

Graduate Research Assistant (AI/ML Engineer)
Quinnipiac University, Hamden, CT, USA

Aug 2024 – Present

- Designed and implemented AI/ML pipelines involving NLP, document understanding, feature engineering, LLM workflows, and experimental analysis using Python and PyTorch
- Conducted AI and Software Engineering research focused on intelligent systems, semantic analysis, AI-generated code evaluation, and machine learning-driven applications
- Collaborated with faculty advisors on experimentation, model evaluation, prompt engineering, technical documentation, and publishable AI/ML research projects

Applied AI Engineer
Uniphore, Chennai, India

Jun 2022 – Apr 2024

- Developed and deployed AI-powered applications using Large Language Models (LLMs), AI agents, NLP, and machine learning techniques, automating enterprise workflows and improving operational efficiency by **30%**.
- Built scalable AI agents and RAG pipelines, reducing response and information retrieval times by **40%**
- Collaborated with cross-functional teams to optimize model performance and production deployments, improving AI system reliability by **25%**

Full-Stack Java Software Engineer
Ijona Technologies Pvt. Ltd, Bangalore, India

Nov 2021 – May 2022

- Developed and validated backend algorithmic workflows using Java and SQL to improve software scalability, reliability, and operational efficiency
- Conducted user behavior and system interaction analysis to optimize application design, document processing workflows, and system performance
- Improved system reliability through debugging, testing, performance analysis, and continuous optimization across software engineering workflows

PROJECTS

LLM-Based Bug Report Enhancement Pipeline | Python, NLP, LLMs

Jan 2026 – Present

- Built an LLM-based NLP pipeline to transform unstructured bug reports into standardized formats, improving report completeness by 72% across 15,000+ GitHub API bug reports
- Developed automated preprocessing and structured information extraction workflows to improve debugging, issue triaging, and semantic analysis

AI-Based Code Generation Evaluation | Python, Java, Published Research

Aug 2025 – Feb 2026

- Conducted empirical evaluation of AI-generated code using 200+ code samples and 1200+ experimental runs across similarity and logical correctness metrics
- Published research at IISec 2026 focused on assessing AI-assisted software engineering and AI-generated code quality evaluation techniques

Perceived Stress Prediction Using Machine Learning with Feature Selection | Python, XGBoost, Scikit-learn

Sep 2025 – May 2026

- Developed machine learning models to predict perceived stress levels using EMA stress and sleep datasets from the StudentLife research study
- Applied feature engineering, mutual information-based feature selection, and cross-validation using XGBoost, SVM, Random Forest, and Neural Networks

Flow Pilot: Enterprise AI Workflow Automation Platform | Next.js, TypeScript, PostgreSQL

Jan 2026 – May 2026

- Built a full-stack enterprise AI workflow automation platform supporting workflow orchestration, approval systems, and API-driven operations
- Developed scalable backend workflows and database-integrated features using Next.js, TypeScript, PostgreSQL, Prisma, and modern engineering practices

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript, TypeScript, C/C++
AI / Machine Learning: PyTorch, Machine Learning, NLP, LLMs, Transformers, Hugging Face, OpenAI APIs, Scikit-learn, XGBoost
Generative AI: Prompt Engineering, AI Agents, RAG Systems, LangChain, Semantic Search, LLM Workflows
Frameworks / Tools: AWS, Docker, Kubernetes, Git, Airflow, MLflow, CI/CD
Software Engineering: REST APIs, Backend Systems, System Design, Agile, SDLC
Concepts: Data Structures & Algorithms, OOP, Scalable Systems, Intelligent Automation

PUBLICATIONS

- Sekar, T., & ElKharboutly, R., “Assessing AI-Based Code Generation Using Code Comments,” 2026 5th International Conference on Information Systems Engineering (IISec), IEEE Xplore, 2026
- Sekar, T., ElKharboutly, R., & Sahoo, S., “Perceived Stress Prediction Using Machine Learning with Feature Selection on EMA Stress and Sleep Data: A StudentLife Study,” at IEEE/ACM CHASE 2026 Poster & Demo Track